

INTERNATIONAL news

TSMC to receive \$6.6B under US CHIPS and Science Act

The US Department of Commerce has finalised a \$6.6 billion funding award under the CHIPS and Science Act to TSMC to support the company's investment in three advanced semiconductor fabrication plants in Phoenix, Arizona, with the first facility expected to commence operations next year. Around 6000 direct employment and more than 20,000 jobs in the construction area will be created from this investment, claims the Commerce Department. In addition to the direct funding, TSMC is eligible for up to \$5 billion in government loans, facilitating its total investment of over \$65 billion in the Arizona facilities.

Beijing plans \$4.6B investment in 12-inch wafer fab

Beijing is set to invest 33 billion yuan (US\$ 4.6 billion) in building a 12-inch wafer fabrication facility, led by state-owned enterprises and funds, as part of China's push to enhance domestic semiconductor production. Beijing Electronics IC Manufacturing, a subsidiary of Beijing Electronics Holding, will oversee the fab project. The project involves companies such as Beijing Yandong Microelectronics (YDME), listed on Shanghai's Star Market, and BOE Technology, a display maker based in China. YDME announced it would invest 4.99 billion yuan in the venture, giving it a controlling 24.95% stake supported by an acting-in-concert agreement.

US plans to stop enhancing China's military intelligence capabilities

The US recently announced that it is finalising existing regulations to limit investments in Chinese technologies, including AI, semiconductors, and quantum systems, to prevent enhancing China's military intelligence capabilities. The rules, effective since January 2024, stem from a 2023 executive order and focus on safeguarding national security by restricting US investments in sectors linked to China's defence advancements.

Wolfspeed secures \$750 million from US government for chip fab



Representational image

Wolfspeed, a US chipmaker, has secured \$750 million through the CHIPS and Science Act to build a new semiconductor fab in North Carolina and expand its New York facility. The projects will create 5000 jobs and

increase silicon carbide wafer production for electric vehicles, AI, and clean energy, with a \$6 billion investment plan.

MosChip joins Renesas as AI/ML design partner

MosChip Technologies has joined Renesas as an AI/ML design partner for the RZ/V series microprocessors. This collaboration aims to deliver advanced AI/ML solutions for automotive, healthcare, and other industries. MosChip will help develop smart safety, patient monitoring, and ADAS solutions, utilising Renesas' power-efficient MPUs for real-time AI processing without cooling requirements.

Intel's \$16.6 billion quarterly loss sparks takeover rumours

Intel reported a \$16.6 billion loss for Q3, its worst in 56 years, due to asset write-downs and a \$2.8 billion restructuring charge from

earlier layoffs. The loss follows struggles with declining profit margins and slow progress in the AI chip market. It has fuelled speculation of a potential takeover as Intel's market value drops below \$100 billion.

Rivals Intel and AMD unite for improved chip compatibility

Intel and AMD, traditionally competitors, have teamed up to enhance the x86 architecture and improve compatibility between their versions. The collaboration, involving industry giants like Google, Microsoft, and IBM, aims to standardise features and simplify development. This initiative could make x86 a more competitive alternative to ARM architecture, with initial results expected by late 2025.

Intel and AIST to build EUV R&D facility in Japan

Intel and Japan's AIST are collaborating to build an R&D facility focused on semiconductor equipment and materials. This centre will introduce extreme ultraviolet lithography technology, which will be the first in Japan. Set to be completed in 3 to 5 years, the facility supports Japan's semiconductor industry growth, which aims to triple by 2030.

Samsung to quit LED business and focus elsewhere

Samsung is exiting the LED business by 2030 due to disappointing financial results and a loss of market competitiveness. The company will phase out LED chip production for lighting, TV backlights, and automotive LEDs by 2026, or latest by 2030. Samsung plans to redirect its focus to more promising areas like power semiconductors and Micro LED technology and reassign affected staff.

Teledyne acquiring Excelitas' defence businesses for \$710 million

Teledyne Technologies will acquire Excelitas' aerospace and defence businesses for \$710 million, with the deal expected to close in early 2025, pending regulatory approval. The acquisition includes Excelitas' optical and advanced electronics systems businesses, adding to Teledyne's defence and space portfolio. The move aims to create synergies and expand Teledyne's customer base.

GlobalFoundries fined \$500,000 for shipment to restricted firm

GlobalFoundries has been fined \$500,000 by the US government for mistakenly shipping \$17 million worth of products to SJ Semiconductor, a restricted Chinese company. The error occurred due to a data glitch in its screening system. SJ Semiconductor is linked to China's SMIC, which the US considers a national security threat. GlobalFoundries voluntarily reported the issue.

ISE Labs expands with new OSAT facility in Mexico

ISE Labs has acquired land in Guadalajara, Jalisco, Mexico, to build a semiconductor assembly and testing (OSAT) facility, expanding its North American operations. The new facility will create over 500 jobs in its first year and support regional economic growth. The project aims to collaborate with local institutions and contribute to Jalisco's dominance in the semiconductor market.

Indonesia blocks Google Pixel sales over local rules

Indonesia has banned the sale of Google Pixel phones for failing to meet local content rules, which require 40% of smartphone components to be sourced domestically. This follows a similar ban on Apple's iPhone 16. Critics argue that the move harms consumers' and investors' confidence, while local market leaders like OPPO and Samsung continue to dominate sales.